Number Sequences with For Loop and Functions

Problem 0 (Display following Sequences)

Write a program which prints all the numbers in the following sequences upto 'N' (asked from the user).

- **1, 3, 5, 7, 9, 11,...**
- **1, 5, 25, 125,...**
- **1, 3, 9, 27, 81,...**
- **1, 4, 8, 11, 22, 25, 50, 53,...**
- **(N), (N-1), (N-2), ..., 3, 2, 1**
- **512(N), 256(N/2), 128,..., 1**

- **1, 1, 3, 6, 5, 11, 7, 16, 9, 21, ...**
- **1, 100, 2, 99, 3, 98, 4, 97, ..., 100, 1**
- **0, 0, 4, 8, 8, 16, 12, 24, 16, 32, ...**
- **10, 20, 15, 22, 20, 24, 25, 26, 30, ...**
- **1, 1, 2, 4, 6, 9, 24, 16, 120, 25, ...**

Problem 1 (Even Odd)

Write a program that takes inputs until the user enters -1 and your program tells the frequency of even and odd numbers.

Problem 2 (Maximum Minimum)

I. Write a program that takes inputs until the user enters -1 and your program tells the maximum and minimum number.

II. Write a program that takes unique integers as inputs until the user enters -1 and your program tells the maximum and 2nd maximum entered till the end.

Problem 3 (Primes)

I. Write a function which takes a number N and tells whether N is a prime number or not (make a function which returns true when N is a prime)

II. Extend I) to take first H, which means it will ask user H inputs and tells about each number whether it is prime or not.

III. Extend I) and write a Program which takes a Number N and tells it's all prime factorization (make a separate function PrintPrimeFactors which prints all the prime a factorization)

IV. Write a program which takes a composite number N and tells its largest prime factor (make a function that returns the largest prime factor)

V. Print all prime numbers between a given range asked from the user
R1-R2: 10 20
The prime number between 10 to 20 are: 11 13 17 19

Problem 4 (Fibonacci Numbers)

I. Write a function which takes as parameter N>=0 and tells (returns) Nth Fibonacci number
e.g. **Fn = Fn-1 + Fn-2, F0 = 0, F1 = 1** http://en.wikipedia.org/wiki/Fibonacci_number
0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55.....

II. Write a program that takes input **'k'** from the user which means the user is interested in **k Fibonacci numbers** and then ask again one by one these k Fibonacci numbers who wants to know. (Reuse part I)

Input Format:

How many Fibonacci Numbers you want to ask: 3
Which Fibonacci: 6
F6= 8
Which Next Fibonacci: 3
F3 = 2
Which last Fibonacci you want to ask is: 4
F4 = 3

III. Write a program which takes as parameter '**T**' and prints all the **Fibonacci numbers less than T**. (Reuse part I)

Sample Input: **You want to Print Up to: 50**
Sample Output: **The Sequence Up to <50 is: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34**

IV. Write a program which takes two parameters **Start** and **End** and prints all the Fibonacci numbers between them. (Reuse part I)

V. By considering the terms in the Fibonacci sequence whose values do not exceed four million, find the sum of the even-valued terms.

Problem 5 (Natural Numbers)

Write a program which adds all the natural numbers below 3000 and greater than 500 that are multiples of 3 or 5 but not both.

Problem 6 (Triangle Numbers)

I. Write a Function which takes as parameter N and returns Nth Triangle Number.
II. Write a program(using ii) which prints the triangle Numbers sequence up to M (asked from the user),

NOTE: The sequence of triangle numbers is generated by adding the natural numbers. So the **7th triangle number would be 1 + 2 + 3 + 4 + 5 + 6 + 7 = 28.**

The first ten terms of triangle number would be: 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, ...

Problem 7 (GCD)

I. Write a function which takes two numbers and returns its Greatest Common Divisor (GCD)

NOTE: GCD of two numbers is the largest number that divides both of them. For Example:

GCD of 20 and 28 is 4 **GCD of 98 and 56 is 14**

II. Write a program that takes a number **n** from the user and then takes **n** inputs from the user and finds their GCD, Reuse the I) part.

Problem 1 (Perfect Square Numbers - for loops with different sentinel conditions)

I. Write a function which tells the integer square root of an integer. For example **for 49 and 50 it should return 7**
II. Write a program that keeps on taking input from the user, until the user enters -1 and tells whether the input number is a complete square or not (It should use Part 1 as a helper function).
III. Modify your program such that until the user enters -1 it keeps on asking Numbers and telling side by side the number is Perfect Square of which number (make a separate function to return this number).
IV. **Modify IsPrime function such that it should not check until sqrt(N).**

Note: **A Complete Square or Perfect Square** is an integer that can be expressed as a product of two equal integers. For example, **9 is a Perfect Square number, since it can be written as 3 × 3**
12 is not a Perfect Square number, since it can not be written as a product of two equal integers.

Problem 2 (Palindrome and Reverse)

I. Write a function that takes an input integer N from the user and makes another variable with M reverse of the number entered and prints it. (write a function which returns M)

Sample Input:	**N = 12345**	**Sample Output:**	**M = 54321**

II. Write a program (function) that takes an integer **N** from the user and tells (returns) whether N is a Palindrome or Not.

Sample Input:	**N = 12321**	**Sample Input:**	**N = 12320**
Sample Output:	**palindrome**	**Sample Output:**	**not palindrome**

Problem 3 (Satisfying Equations Problems)

I. Write a Program which prints all the triplet a, b and c which satisfies this: **a+b+c = 1000**

II. Write a Program which finds a triplet a, b and c (3 integers) whose sum satisfies this: $\mathbf{a^2+b^2+c^2 = 1000}$

III. Find three consecutive numbers whose multiplication is equal to the Required Number, (**Your program should prompt for the number N and output 3-numbers those if we multiply, equals to N. Also if there isn't any 3-tuple of numbers then it should say NO**).

IV. Print all pythagorean triples upto the given range **R**.

Sample Input:	**R = 20**
Sample Output:	**(3,4,5), (6,8,10), (5,12,13), (9,12,15), (8,15,17), (12, 16, 20)**

Homework

Problem 1 (Decimal to Binary Conversion)

Write a program (function) that takes **N** in decimal and outputs its binary representation.

Sample Input:	**N = 2**	**Sample Input:**	**N = 25**
Sample Output:	**10**	**Sample Output:**	**11001**

Problem 2 (Factorial)

I. Write a function that returns the factorial of a number.
Note: The factorial of a number is the product of all the integers from 1 to that number. For example, the **factorial of 6, 6! = 1*2*3*4*5*6 = 720**. Factorial is not defined for negative numbers, and the **factorial of zero is one, 0! = 1**

II. Write a function which will take two inputs and find nPr (number of permutations). [Use part i)]
Note: nPr(n,r) = n! / (n-r)!

III. Write a function which will take two inputs and find nCr(number of combinations) where
nCr(n,r) = n! / (n-r)! * r! [Use part i)]
BONUS: Write a function which will take two inputs and find nCr(number of combinations) where
nCr(n,r) = nPr(n,r) / r! [Use part i and part ii)]

IV. Write a Test Program **main()** for i), ii) and iii)

Nested Loop and Functions (II)

Shapes Printing

Classwork

Problem 1 (Right-angled Triangle)

I. Write a function **PrintASymbolKTimes(char S, int K)**. This prints the symbol S on the screen K times.

II. Using the above functions your program should print the following 4 shapes:

```
(A)          (B)          (C)          (D)
*            **********   **********            *
**           *********     *********           **
***          ********       ********          ***
****         *******         *******         ****
*****        ******           ******        *****
******       *****             *****       ******
*******      ****               ****      *******
********     ***                 ***     ********
*********    **                   **    *********
**********   *                     *   **********
```

Sample Output: Your program first should prompt for the Height of the triangle and which type of the triangle and then print the right angled triangle corresponding to the option entered by the user.

<u>Problem 2 (Equilateral Triangle - Filled and Hollow)</u>

Write a program which takes input from the user, the height **H** and then prints the **filled and hollow** equilateral triangle shape corresponding to the entered input.

Sample Input: H = 6

Sample Output:

```
     *          *
    * *        *   *
   * * *      *     *
  * * * *    *       *
 * * * * *  *         *
* * * * * * * * * * * * *
```

<u>Problem 3 (Diamond - Filled and Hollow)</u>

Write a program which takes input from the user, the height **H** of the Diamond and then prints the diamond shape (**filled and hollow**) corresponding to the entered input.

Sample Input: H = 4

Sample Output:

```
     *          *
    * *        *   *
   * * *      *     *
  * * * *    *       *
 * * * * *    *     *
  * * * *      *   *
   * * *        * *
    *            *
```

Shapes Printing

Homework

Problem 1 (Pajamas)

Print the following shapes:

Printing Pajama ☺

Write a Program which asks for 4 options menu:

Option 1: It Takes the Dimension and prints the following shape:

```
**********  **********
*********    *********
********      ********
*******        *******
******          ******
*****            *****
****              ****
***                ***
**                  **
*                    *
```

Option 2: It takes as parameter and prints the following Shape:

```
*                    *
**                  **
***                ***
****              ****
*****            *****
******          ******
*******        *******
********      ********
*********    *********
**********  **********
```

Option 3: It takes the appropriate parameters and Prints the following shape:

```
**********  **********
*********    *********
********      ********
*******        *******
******          ******
*****            *****
****              ****
***                ***
**                  **
*                    *
*                    *
**                  **
***                ***
****              ****
*****            *****
******          ******
*******        *******
********      ********
*********    *********
**********  **********
```

option 4: Exit

your program should keep on asking until the user Selects Exit Option

BONUS (Number Triangle)

Develop a program which takes the height **H** of a triangle as input and print the triangles in the following fashion.

Note: You should make 4 functions

1. For printing spaces
2. For printing sequence of numbers in ascending order (from 1 to N)
3. For printing sequence of numbers in descending order (from N to 1)
4. For printing complete shape by calling the above mentioned functions

Sample Input : H = 5

Sample Output :

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

```
1 2 3 4 5
1 2 3 4
1 2 3
1 2
1
```

```
A
B B
C C C
D D D D
E E E E E
```

```
        1
      2 3 2
    3 4 5 4 3
  4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5
```

```
1
2 3
4 5 6
7 8 9 10
```

CHALLENGE

Challenge 1

Write a function which takes a number N and tells whether it is a circular prime or not. Read about the circular prime here https://en.m.wikipedia.org/wiki/Circular_prime

Challenge 2

Detect whether a number is a Truncatable prime or not.
https://en.m.wikipedia.org/wiki/Truncatable_prime

Challenge 3

Goldberg's conjecture is a famous mathematical statement which says the following any even number can be written as the sum of two prime numbers, write a program which first ask for a number K which means the user can give K even integers greater than 4 iteratively (in loop) and your program should display the two primes for every entered number which sum to the given number.

Challenge 4

Write a program which takes a range (two numbers R1 and R2) and report all twin primes between the given range. Twin primes are prime numbers which are consecutive odd numbers like 17, 19 similarly 29, 31 a d 41, 43.

```
O===O
 \O/
  |
```

/\

Happy Coding... :)

Good luck